

Helix OTA Server — Real Build/ Test Health Baseline

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Captured per anti-bluff/no-guessing discipline — every command below was actually executed in `/home/milos/Factory/projects/tools_and_research/helix_ota/server`, and every exit code / output shown is verbatim from that execution. No source files were modified. No git add/commit/push was performed.

1. Go version

Command: `go version`

```
go version go1.26.4-X:nodwarf5 linux/amd64
```

2. go build ./...

Command (from server/):

```
go build ./... > /tmp/build_out.txt 2>&1; echo "BUILD_EXIT=$?"  
>> /tmp/build_out.txt
```

Output:

```
BUILD_EXIT=0
```

(No compiler output preceded the exit line — a clean build.)

Result: BUILD OK (exit 0).

3. go vet ./...

Command:

```
go vet ./... > /tmp/vet_out.txt 2>&1; echo "VET_EXIT=$?" >> /tmp/
vet_out.txt
```

Output:

```
VET_EXIT=0
```

(No vet diagnostics preceded the exit line.)

Result: VET OK (exit 0).

4. **gofmt -l .**

Command:

```
gofmt -l . > /tmp/gofmt_out.txt 2>&1; echo "GOFMT_EXIT=$?" >> /
tmp/gofmt_out.txt
```

Output:

```
GOFMT_EXIT=0
```

(No filenames listed — gofmt reported zero formatting violations. gofmt -l itself always exits 0 regardless of findings; the empty file list before the exit line is what confirms a clean tree.)

Result: GOFMT CLEAN — no files listed.

5. **timeout 300 go test ./...**

Command:

```
timeout 300 go test ./... > /tmp/test_out.txt 2>&1; echo
"TEST_EXIT=$?" >> /tmp/test_out.txt
```

Full verbatim output:

```
ok      github.com/HelixDevelopment/helix_ota/server/cmd/
applyport 0.975s
ok      github.com/HelixDevelopment/helix_ota/server/cmd/ota-
device-emu 0.961s
ok      github.com/HelixDevelopment/helix_ota/server/cmd/ota-
server 2.495s
ok      github.com/HelixDevelopment/helix_ota/server/internal/
api 0.382s
```

```

?      github.com/HelixDevelopment/helix_ota/server/internal/api/
manager-dist [no test files]
ok      github.com/HelixDevelopment/helix_ota/server/internal/
config  0.002s
ok      github.com/HelixDevelopment/helix_ota/server/internal/
device  0.024s
ok      github.com/HelixDevelopment/helix_ota/server/internal/
deviceemu 0.282s
ok      github.com/HelixDevelopment/helix_ota/server/internal/
fabric  0.004s
ok      github.com/HelixDevelopment/helix_ota/server/internal/
health  0.002s
ok      github.com/HelixDevelopment/helix_ota/server/internal/
rollout 0.004s
ok      github.com/HelixDevelopment/helix_ota/server/internal/
store 0.023s
ok      github.com/HelixDevelopment/helix_ota/server/internal/
transport 0.112s
ok      github.com/HelixDevelopment/helix_ota/server/tests/
chaos    0.026s
ok      github.com/HelixDevelopment/helix_ota/server/tests/
stress  0.010s
ok      github.com/HelixDevelopment/helix_ota/server/tools/
loadtest 1.404s
TEST_EXIT=0

```

No hang, no timeout — full suite completed in well under the 300s budget (slowest package cmd/ota-server at 2.495s).

Per-package result:

Package	Result	Time
cmd/applyport	ok	0.975s
cmd/ota-device-emu	ok	0.961s
cmd/ota-server	ok	2.495s
internal/api	ok	0.382s
internal/api/manager-dist	no test files	—
internal/config	ok	0.002s
internal/device	ok	0.024s
internal/deviceemu	ok	0.282s
internal/fabric	ok	0.004s
internal/health	ok	0.002s

Package	Result	Time
internal/rollout	ok	0.004s
internal/store	ok	0.023s
internal/transport	ok	0.112s
tests/chaos	ok	0.026s
tests/stress	ok	0.010s
tools/loadtest	ok	1.404s

Result: 15/15 tested packages PASS, 0 FAIL, 1 package (internal/api/manager-dist) has no test files (not integration-gated — genuinely empty of `_test.go` files; a coverage gap, not a failure). No package required `-tags integration` or external `infra` to run — the full module tree ran under the default `go test ./...` invocation.

6. timeout 300 go test -cover ./...

Command:

```
timeout 300 go test -cover ./... > /tmp/cover_out.txt 2>&1; echo
"COVER_EXIT=$?" >> /tmp/cover_out.txt
```

Full verbatim output:

```
ok      github.com/HelixDevelopment/helix_ota/server/cmd/
applyport 1.437s coverage: 24.4% of statements
ok      github.com/HelixDevelopment/helix_ota/server/cmd/ota-
device-emu 1.444s coverage: 0.0% of statements
ok      github.com/HelixDevelopment/helix_ota/server/cmd/ota-
server 1.867s coverage: 5.1% of statements
ok      github.com/HelixDevelopment/helix_ota/server/internal/
api 0.158s coverage: 92.4% of statements
?       github.com/HelixDevelopment/helix_ota/server/internal/api/
manager-dist [no test files]
ok      github.com/HelixDevelopment/helix_ota/server/internal/
config 0.002s coverage: 100.0% of statements
ok      github.com/HelixDevelopment/helix_ota/server/internal/
device 0.042s coverage: 92.6% of statements
ok      github.com/HelixDevelopment/helix_ota/server/internal/
deviceemu 0.265s coverage: 97.7% of statements
ok      github.com/HelixDevelopment/helix_ota/server/internal/
fabric 0.004s coverage: 100.0% of statements
ok      github.com/HelixDevelopment/helix_ota/server/internal/
```

```
health    0.002s  coverage: 100.0% of statements
ok        github.com/HelixDevelopment/helix_ota/server/internal/
rollout   0.004s  coverage: 28.2% of statements
ok        github.com/HelixDevelopment/helix_ota/server/internal/
store     0.005s  coverage: 47.9% of statements
ok        github.com/HelixDevelopment/helix_ota/server/internal/
transport 0.110s  coverage: 100.0% of statements
ok        github.com/HelixDevelopment/helix_ota/server/tests/
chaos     0.025s  coverage: [no statements]
ok        github.com/HelixDevelopment/helix_ota/server/tests/
stress    0.010s  coverage: [no statements]
ok        github.com/HelixDevelopment/helix_ota/server/tools/
loadtest  1.383s  coverage: 63.3% of statements
COVER_EXIT=0
```

Per-package coverage:

Package	Coverage
cmd/applyport	24.4%
cmd/ota-device-emu	0.0%
cmd/ota-server	5.1%
internal/api	92.4%
internal/api/manager-dist	no test files
internal/config	100.0%
internal/device	92.6%
internal/deviceemu	97.7%
internal/fabric	100.0%
internal/health	100.0%
internal/rollout	28.2%
internal/store	47.9%
internal/transport	100.0%
tests/chaos	no statements (test-only package)
tests/stress	no statements (test-only package)
tools/loadtest	63.3%

Honest gap observations (stated as fact, not speculation): - cmd/ota-device-emu (0.0%), cmd/ota-server (5.1%), cmd/applyport (24.4%) are main-package command entry points — low coverage here is typical for main.go wiring code exercised mainly through integration/e2e paths rather than unit tests; this is a coverage gap, not a build/test failure. -

internal/rollout (28.2%) and internal/store (47.9%) are the two lowest non-main-package coverage figures and are candidates for additional unit test investment. - internal/api/manager-dist has zero test files — flagged as a coverage gap requiring an honest tracked item, not a silent absence.

Summary

Check	Result
Go version	go1.26.4-X:nodwarf5 linux/amd64
go build ./...	OK (exit 0)
go vet ./...	OK (exit 0)
gofmt -l .	CLEAN (no files listed)
go test ./...	PASS — 15/16 packages ok, 1 no test files, exit 0
go test -cover ./...	exit 0; coverage ranges 0.0%–100.0% per package (see table)

No hangs, no timeouts, no failures encountered during this baseline capture.